Federated Learning

A distributed ML approach where multiple clients collaboratively train a shared global model without sharing their raw local data.

→ Privacy
→ Data locality

1) Server initializes a global model and send it to selected clients.
2) Each clients trains the model locally on its own data.
3) Clients send only their updated model weights back to server.
4) Server aggregates these updates to form a new global model.
5) Repeat for multiple communication round until convergence.

FedAvg → The most common aggregation algorithm. The server computes a weighted avg of client model updates typically weighted by the no of local data samples each client has.

$$w\_global = \Sigma(n_k/n) \cdot w_k$$

Decentralized Federated Learning →
No central server - Clients communicate directly with peers. (often in network/graph topology) exchanging and aggregating model updates

among themselves. Removing the single point of failure/bottleneck of central server.

Lightweight Model

A model designed to have a small memory footprint low computational cost & fast inference optimized for deployment on resource constraint device while trying to retain resonable accuracy.

we need because edge device need limited compute, memory, battery and no reliable internet connectivity for cloud inference.

MobileNet

Build around depthwise seperable convolutions instead of standard convolution

drastically cutting parameters/FLOPs

Standard Convolution → Depthwise Convolution (single filter/input independent)

→ Pointwise Convolution (combines the o/p across channel)

Raspberry Pi → A small, low cost, single board computer capable of running a full OS, commonly used in ICT, embedded AI.

Deploying AI on Raspberry – Enable real time on device inference without needing constant cloud connectivity reducing latency, bandwidth usage and dependency on internet while improving data privacy.